Backyard Toolshed Routine

Week 7 Assignment - Problem 3 (Intermediate)

The hierarchy is GardenTool -> CuttingTool -> Pruner. Each subclass constructor calls super(), and each override layers new detail onto the parent's result.

Java solution

```
abstract class GardenTool {
    // Concrete base method is needed for super.use() to compile.
    public String use() { return ""; }
}
class CuttingTool extends GardenTool {
    CuttingTool() { super(); }
    @Override public String use() {
        return super.use()
            + "Using the tool in the garden, blade sharpened first";
    }
}
class Pruner extends CuttingTool {
    Pruner() { super(); }
    @Override public String use() {
        return super.use() + ", then trimming branches precisely";
    }
}
public class Main {
    public static void main(String[] args) {
        CuttingTool c = new CuttingTool();
        Pruner p = new Pruner();
        System.out.println(c.use());
        System.out.println(p.use());
    }
}
```

Sample output

```
Using the tool in the garden, blade sharpened first
Using the tool in the garden, blade sharpened first, then trimming branches precisely
```

Note: Source conflict: GardenTool.use() is requested as abstract, but CuttingTool is also told to call super.use(). Java cannot call an abstract method with super. To keep the requested super-call chain compilable, GardenTool is abstract but provides a concrete empty base use() method.